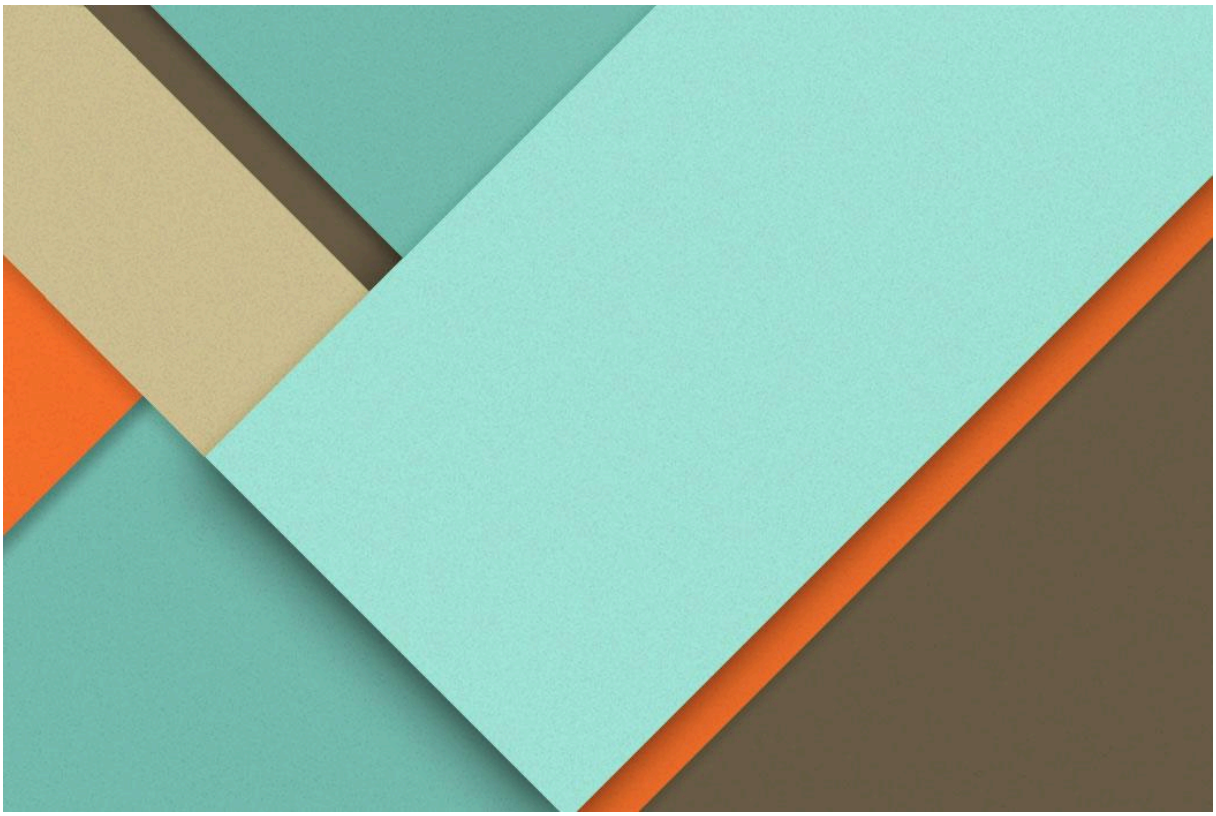


Dreamxec-Phase 2

09.03.2026

PRD

It translates the **Product Vision + Architecture Map + Capability Map** into **clear modules, feature specifications, workflows, user stories, and acceptance criteria**.

This PRD intentionally goes deep and covers **150+ features** so engineering team can build the system without major gaps. (We need to still scout for gaps)

Product Requirements Document (PRD)

Product: DreamXec
Phase: Phase-2
Document Type: Product Requirements Document
Version: 1.0
Prepared For: DreamXec Product & Engineering Teams

1. Product Overview

DreamXec Phase-2 transforms the platform into a **complete Research ERP and Innovation Ecosystem,** enabling organizations to manage the entire lifecycle of research and innovation.

The platform will support:

- Research idea generation
- Project lifecycle management
- Cross-institution collaboration
- Shared research infrastructure
- Grant and funding management
- Compliance workflows
- Intellectual property tracking
- Knowledge management
- Research commercialization

DreamXec acts as a **multi-organization research operating system** connecting universities, corporations, incubators, government agencies, and research professionals.

2. Product Objectives

Primary objectives of Phase-2:

1. Provide **complete research lifecycle management**
2. Enable **cross-organization collaboration**
3. Improve **research funding transparency**
4. Optimize **research resource utilization**
5. Manage **research knowledge and IP**
6. Ensure **regulatory compliance**
7. Support **research commercialization pipelines**

3. User Personas

Primary users:

• Researchers
• Professors
• Corporate R&D managers
• Students
• Grant officers
• Incubator managers
• Mentors
• Compliance officers
• Platform administrators

Each persona interacts with specific modules.

4. Product Modules

DreamXec Phase-2 consists of **10 major modules**:

1. Identity & Access Management
2. Organization & Ecosystem Management
3. Idea & Proposal Management
4. Project & Portfolio Management
5. Resource & Infrastructure Management
6. Funding & Grant Management
7. Knowledge & Research Data Management
8. Intellectual Property Management
9. Compliance & Governance
10. Collaboration & Communication

MODULE 1

Identity & Access Management

Purpose: Secure authentication and multi-organization access control.

Features

IAM-001

User Registration

Users can create accounts with:

• email
• institution
• role
• research interests

Acceptance Criteria

User receives verification email.

IAM-002

User Authentication

Support login using:

• email/password
• SSO
• OAuth

IAM-003

Multi-Factor Authentication

Users can enable MFA using:

• authenticator apps
• SMS verification

IAM-004

Password Management

Users can:

• change password
• reset password
• enforce password policy

IAM-005

Session Management

System tracks active sessions and allows logout from all devices.

IAM-006

Single Sign-On

Support SAML and OAuth identity providers.

IAM-007

LDAP Integration

Universities can connect institutional directories.

IAM-008

Role Management

Roles include:

- researcher
- professor
- mentor
- corporate admin
- compliance officer

IAM-009

Permission Management

Permissions can be assigned per:

- organization
- project
- resource

IAM-010

Attribute-Based Access

Access policies can include attributes like:

- organization
- department
- project membership

IAM-011

User Profile

User profiles store:

- research areas
- publications

• expertise
• certifications

IAM-012

Account Deactivation

Admins can deactivate users.

MODULE 2

Organization & Ecosystem Management

Purpose: Manage institutions and partnerships.

ORG-001

Organization Registration

Organizations can register as:

• university
• corporate R&D
• incubator
• research lab
• funding agency

ORG-002

Organization Verification

Admin verifies institutional legitimacy.

ORG-003

Department Management

Organizations can create departments.

ORG-004

Research Group Management

Departments can create research groups.

ORG-005

Lab Management

Labs can be registered with the equipment inventory.

ORG-006

Organization Directory

Public directory of research organizations.

ORG-007

Partnership Management

Organizations can establish research partnerships.

ORG-008

Collaboration Agreements

Digital agreements between institutions.

ORG-009

External Collaborators

Invite external researchers.

ORG-010

Institution Profiles

Profiles include:

• research focus
• labs
• collaborations

ORG-011

Organization Analytics

Shows:

• research output
• collaborations
• funding received

MODULE 3

Idea & Proposal Management

Purpose: Capture innovation ideas.

IDEA-001

Idea Submission

Users submit research ideas.

Fields include:

• title
• description
• research area
• expected impact

IDEA-002

Idea Tagging

Ideas can be tagged with:

• TRL level
• SDG goals
• technology domains

IDEA-003

Idea Review Workflow

Ideas are reviewed by:

• mentors
• incubators
• organizations

IDEA-004

Idea Scoring

Ideas scored based on:

- novelty
- feasibility
- impact

IDEA-005

Idea Feedback

Reviewers can provide feedback.

IDEA-006

Idea Collaboration

Users can request collaborators.

IDEA-007

Idea Conversion

Ideas can be converted to research projects.

IDEA-008

Idea Versioning

Idea revisions are tracked.

IDEA-009

Idea Search

Users can search ideas.

IDEA-010

Idea Dashboard

Shows idea pipeline.

MODULE 4

Project & Portfolio Management

Core research management engine.

PROJ-001

Project Creation

Projects created from ideas or directly.

PROJ-002

Project Templates

Templates for common research workflows.

PROJ-003

Project Stages

Stages include:

- discovery
- feasibility
- prototype
- pilot
- commercialization

PROJ-004

Stage Gate Approvals

Each stage requires approvals.

PROJ-005

Task Management

Create tasks and assign owners.

PROJ-006

Milestone Tracking

Track key deliverables.

PROJ-007

Project Dependencies

Projects can depend on others.

PROJ-008

Project Teams

Assign team members.

PROJ-009

Cross Organization Projects

Projects may include multiple institutions.

PROJ-010

Project Timeline

Gantt view of project schedule.

PROJ-011

Project Budget Integration

Connect project to budget.

PROJ-012

Project Analytics

Track performance metrics.

PROJ-013

Project Risk Tracking

Identify risks.

PROJ-014

Go/No-Go Decisions

Capture decision outcomes.

PROJ-015

Portfolio Dashboard

View all projects.

MODULE 5

Resource & Infrastructure Management

RES-001

Equipment Registration

Register equipment.

RES-002

Equipment Metadata

Track model, location, specifications.

RES-003

Equipment Booking

Researchers reserve equipment.

RES-004

Booking Calendar

View availability.

RES-005

Conflict Detection

Prevent double booking.

RES-006

Equipment Maintenance

Schedule maintenance.

RES-007

Lab Capacity Monitoring

Track lab utilization.

RES-008

Resource Cost Tracking

Track usage cost.

RES-009

Researcher Skill Profiles

Store expertise.

RES-010

Skill Matching

Suggest team members.

RES-011

Resource Utilization Analytics

Analyze usage trends.

RES-012

Equipment Sharing

Allow cross-institution usage.

MODULE 6

Funding & Grant Management

GRANT-001

Grant Listing

Funding agencies list grants.

GRANT-002

Grant Discovery

Search grants.

GRANT-003

Grant Matching

Match projects to grants.

GRANT-004

Grant Application

Submit proposals.

GRANT-005

Budget Planning

Prepare budgets.

GRANT-006

Institutional Approval

Organizations approve submissions.

GRANT-007

Grant Allocation

Assign funds to projects.

GRANT-008

Grant Budget Monitoring

Track expenses.

GRANT-009

Grant Reporting

Generate reports.

GRANT-010

Milestone Based Funding

Release funds upon milestones.

GRANT-011

Funding Analytics

Track funding distribution.

GRANT-012

Financial Integration

Export financial reports.

MODULE 7

Knowledge & Research Data Management

KNOW-001

Research Repository

Central storage.

KNOW-002

Dataset Management

Store datasets.

KNOW-003

Document Management

Upload papers and reports.

KNOW-004

Electronic Lab Notebook

Log experiments.

KNOW-005

Experiment Timestamping

Record timestamps.

KNOW-006

Research Version Control

Track revisions.

KNOW-007

Metadata Indexing

Tag research data.

KNOW-008

Semantic Search

AI-based search.

KNOW-009

Data Access Control

Restrict data access.

KNOW-010

Research Analytics

Analyze research outputs.

MODULE 8

Intellectual Property Management

IP-001

Invention Disclosure

Submit invention reports.

IP-002

Inventor Attribution

Track contributors.

IP-003

Patent Drafting

Prepare patent documents.

IP-004

Patent Filing Tracking

Monitor filings.

IP-005

Patent Status Monitoring

Track application status.

IP-006

Renewal Tracking

Track renewals.

IP-007

Joint Ownership

Track ownership shares.

IP-008

Licensing Agreements

Manage licenses.

IP-009

Royalty Tracking

Track revenues.

IP-010

IP Analytics

Monitor innovation metrics.

MODULE 9

Compliance & Governance

COMP-001

IRB Protocol Submission

Submit ethics applications.

COMP-002

Ethics Committee Review

Review protocols.

COMP-003

Approval Workflow

Approve protocols.

COMP-004

Renewal Tracking

Track renewals.

COMP-005

Regulatory Compliance

Track standards.

COMP-006

Agreement Management

Track legal agreements.

COMP-007

Audit Trails

Record system actions.

COMP-008

Compliance Reporting

Generate reports.

COMP-009

Risk Monitoring

Identify compliance risks.

COMP-010

Incident Reporting

Report violations.

MODULE 10

Collaboration & Communication

COL-001

Project Messaging

Chat within projects.

COL-002

Direct Messaging

User messaging.

COL-003

Discussion Threads

Project discussions.

COL-004

Shared Documents

Collaborative documents.

COL-005

Mentor Dashboard

Track mentee progress.

COL-006

Mentor Feedback

Provide feedback.

COL-007

Recognition System

Award badges.

COL-008

Innovation Leaderboards

Rank contributors.

COL-009

Event Announcements

Share events.

COL-010

Meeting Integration

Video meeting links.

Feature Count Summary

Approximate totals:

- Identity & Access → 12
- Organization → 11
- Ideas → 10
- Projects → 15
- Resources → 12
- Grants → 12
- Knowledge → 10
- IP → 10
- Compliance → 10
- Collaboration → 10

Total Features ≈ **112 core**

When expanded into sub-features and workflows → **150+ capabilities**

Success Metrics

Success indicators include:

- number of research projects
- funding processed
- cross-institution collaborations
- patents generated

End of PRD

DREAMXEC

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Phase-2: Research ERP & Innovation Ecosystem

Version: 1.1
Prepared by: Product Team
Product: DreamXec
Date: 2026

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1. Product Overview

DreamXec Phase-2 transforms the platform from a **crowdfunding system for student projects** into a **full research innovation operating system** connecting universities, corporations, incubators, donors, and researchers.

Phase-2 introduces:

- research project lifecycle management
- shared infrastructure networks
- grant & funding management
- intellectual property management
- research knowledge repository
- compliance & governance workflows
- cross-institution collaboration

The platform becomes a **multi-organization ecosystem for innovation and research commercialization**.

2. Product Goals

Goal 1

Enable end-to-end **research lifecycle management**.

Goal 2

Create a **global research collaboration ecosystem**.

Goal 3

Improve **research funding transparency**.

Goal 4

Enable **research commercialization and IP management**.

Goal 5

Create **shared research infrastructure networks**.

3. Product Scope

Phase 2 expands DreamXec into **10 major systems**:

1 Identity & Access Management
2 Organization & Ecosystem Management
3 Idea & Proposal Management
4 Research Project Management
5 Resource & Infrastructure Management
6 Funding & Grant Management
7 Knowledge & Research Data Management
8 Intellectual Property Management
9 Compliance & Governance
10 Collaboration & Communication

4. User Personas

Researcher / Student

Creates research ideas and projects.

Needs:
• funding
• collaborators
• equipment access

Professor

Mentors students and supervises research.

Needs:
• project oversight
• research data access

Corporate R&D Manager

Sponsors research and monitors outcomes.

Needs:
• portfolio dashboards
• IP ownership tracking

Donor / Sponsor

Funds projects.

Needs:
• project discovery
• transparent progress tracking

Incubator Manager

Evaluates ideas and converts them into startups.

Needs:
• innovation scoring
• commercialization tracking

Compliance Officer

Ensures regulatory compliance.

Needs:
• audit trails
• protocol approvals

5. Identity & Access Management

Feature IAM-1

User registration

Users can register with:

• email
• Google OAuth
• LinkedIn OAuth

Feature IAM-2

Email verification

Users must verify email before platform access.

Feature IAM-3

Role management

Supported roles:

USER
DONOR
ADMIN
STUDENT_PRESIDENT
RESEARCHER
MENTOR
INCUBATOR_MANAGER
COMPLIANCE_OFFICER

Feature IAM-4

Account status management

Account states:

ACTIVE
BLOCKED
SUSPENDED
UNDER_REVIEW

Feature IAM-5

Multi-factor authentication

Users can enable 2FA.

Feature IAM-6

Session management

Users can manage active sessions.

Feature IAM-7

Organization access control

Permissions scoped by:

- organization
- project
- resource

6. Organization & Ecosystem Management

Phase-2 introduces **organizations as first-class entities**.

Feature ORG-1

Organization registration

Organizations can register as:

- university
- corporate R&D
- incubator
- research lab
- funding agency

Feature ORG-2

Organization verification

Admins verify institutions.

Feature ORG-3

Department management

Organizations can create departments.

Feature ORG-4

Research group management

Departments can create research groups.

Feature ORG-5

Lab registration

Labs can be registered with equipment inventory.

Feature ORG-6

Partnership management

Organizations can create collaboration agreements.

Feature ORG-7

External collaborator invitations

Invite researchers outside the organization.

Feature ORG-8

Organization analytics dashboard

Shows:

- research output
- collaborations
- funding received

---

7. Idea & Proposal Management

DreamXec becomes a **global idea pipeline**.

Feature IDEA-1

Idea submission

Users submit research ideas.

Fields:

- title
- description
- research domain
- expected impact

Feature IDEA-2

Idea tagging

Ideas tagged with:

- TRL level
- SDG goals
- technology domains

Feature IDEA-3

Idea review workflow

Reviewers include:

- mentors
- incubators
- admins

Feature IDEA-4

Idea scoring

Ideas scored on:

- novelty
- feasibility
- impact

Feature IDEA-5

Idea feedback

Reviewers provide structured feedback.

Feature IDEA-6

Idea collaboration requests

Users can request collaborators.

Feature IDEA-7

Idea conversion

Approved ideas convert to research projects.

8. Research Project Management

This is the **core of Phase-2**.

Feature PROJ-1

Project creation

Projects can be created from ideas or directly.

Feature PROJ-2

Project stages

Stages include:

Discovery
Feasibility
Prototype
Pilot
Commercialization

Feature PROJ-3

Stage gate approvals

Projects must pass stage approvals.

Feature PROJ-4

Project team management

Assign researchers and collaborators.

Feature PROJ-5

Task management

Create tasks and assign responsibilities.

Feature PROJ-6

Milestone management

Milestones linked to funding release.

Feature PROJ-7

Project timeline

Gantt-style timeline.

Feature PROJ-8

Cross-organization projects

Projects can involve multiple organizations.

Feature PROJ-9

Project risk tracking

Identify project risks.

Feature PROJ-10

Project analytics dashboard

Shows:

- milestone completion
- funding utilization
- collaboration network

---

9. Resource & Infrastructure Management

DreamXec introduces **shared research infrastructure networks**.

Feature RES-1

Equipment registration

Labs register equipment.

Feature RES-2

Equipment metadata

Store:

- model
- specifications
- location

Feature RES-3

Equipment booking

Researchers reserve equipment slots.

Feature RES-4

Booking calendar

View equipment availability.

Feature RES-5

Conflict detection

Prevent double booking.

Feature RES-6

Equipment maintenance tracking

Schedule maintenance.

Feature RES-7

Lab capacity monitoring

Track lab usage.

Feature RES-8

Resource cost tracking

Track usage costs.

Feature RES-9

Researcher expertise database

Profiles include:

- skills
- publications
- certifications

Feature RES-10

AI team recommendations

Suggest collaborators based on skills.

10. Funding & Grant Management

DreamXec introduces **grant lifecycle management**.

Feature GRANT-1

Grant listing

Funding agencies list grants.

Feature GRANT-2

Grant discovery

Users search grants.

Feature GRANT-3

Grant matching

AI suggests grants for projects.

Feature GRANT-4

Grant application

Users submit proposals.

Feature GRANT-5

Budget planning

Prepare grant budgets.

Feature GRANT-6

Institutional approval

Organizations approve submissions.

Feature GRANT-7

Grant allocation

Assign funds to projects.

Feature GRANT-8

Grant reporting

Generate compliance reports.

Feature GRANT-9

Milestone-based funding

Funds released on milestone completion.

Feature GRANT-10

Funding analytics

Track funding distribution.

---

11. Knowledge & Research Data Management

DreamXec becomes a **research knowledge repository**.

Feature KNOW-1

Research document repository

Store papers and reports.

Feature KNOW-2

Dataset storage

Store datasets.

Feature KNOW-3

Electronic lab notebook

Log experiments.

Feature KNOW-4

Experiment timestamping

Ensure data integrity.

Feature KNOW-5

Research version control

Track document revisions.

Feature KNOW-6

Semantic search

Search research data.

Feature KNOW-7

Research metadata indexing

Tag datasets and documents.

---

12. Intellectual Property Management

Feature IP-1

Invention disclosure

Researchers submit inventions.

Feature IP-2

Inventor attribution

Track inventors.

Feature IP-3

Patent drafting workflow

Prepare patent applications.

Feature IP-4

Patent filing tracking

Track filings.

Feature IP-5

Patent status monitoring

Monitor application status.

Feature IP-6

Renewal tracking

Track patent renewals.

Feature IP-7

Joint ownership management

Track IP ownership percentages.

Feature IP-8

Licensing agreements

Manage IP licensing.

Feature IP-9

Royalty tracking

Track IP revenue.

13. Compliance & Governance

Feature COMP-1

IRB protocol submission

Submit ethics protocols.

Feature COMP-2

Ethics review workflow

Committees review submissions.

Feature COMP-3

Protocol approvals

Approve or reject protocols.

Feature COMP-4

Renewal tracking

Track protocol renewals.

Feature COMP-5

Compliance reporting

Generate compliance reports.

Feature COMP-6

Audit trails

Track system actions.

Feature COMP-7

Risk monitoring

Detect compliance risks.

Feature COMP-8

Incident reporting

Report research violations.

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14. Collaboration & Communication

Feature COL-1

Project messaging

Chat within projects.

Feature COL-2

Direct messaging

User messaging.

Feature COL-3

Discussion threads

Threaded discussions.

Feature COL-4

Shared documents

Collaborative editing.

Feature COL-5

Mentor dashboard

Track mentee progress.

Feature COL-6

Mentor feedback

Provide structured feedback.

Feature COL-7

Recognition system

Award innovation badges.

Feature COL-8

Innovation leaderboards

Rank contributors.

Feature COL-9

Event announcements

Share research events.

Feature COL-10

Video meeting integration

Integrate video meetings.

---

15. Non-Functional Requirements

Performance

Support:

100k users
10k concurrent sessions

Security

Implement:

data encryption
access logging
IP protection

Availability

Platform uptime:

99.9%

Scalability

Support horizontal scaling via microservices.

---

16. Success Metrics

Key KPIs:

- number of research projects
- total funding processed
- cross-institution collaborations
- patents generated
- equipment utilization

---

17. Risks

Major risks include:

- data privacy issues
- regulatory complexity
- IP ownership disputes
- institutional adoption barriers

End of PRD
